SUMMARY TABLE: BIOLOGY GRADE 10

Sub-Strand 3.1: Animal Nutrition — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 3.1: Animal Nutrition
Driving Question	DRIVING QUESTION: How do the structures animals use to get food — from an aphid's needle-like mouthparts to a flamingo's filter beak to your own teeth and digestive system — explain how each animal gets the nutrients it needs to survive?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **What Is Animal Nutrition and Why Does It Matter? — Launching the Anchoring Phenomenon**	Students examined a set of anchor images: an aphid feeding on a maize stem in a Rift Valley farm, a flamingo filtering water at Lake Nakuru, and a Maasai elder chewing nyama choma. Teacher should display all three images simultaneously and ask: 'What is the same about what each animal is doing? What is different?' Allow 60 seconds silent observation before cold-calling 4–5 students. Students also observed a short video clip of a locust swarm devastating a maize crop in Western Kenya, anchoring the real-world stakes of animal feeding structures.	Animal nutrition is the process by which animals obtain, digest, absorb, and use food for energy, growth, repair, and reproduction. All animals are heterotrophs — they cannot manufacture their own food and must consume organic matter. Different animals have evolved radically different feeding structures (mouthparts, beaks, teeth, digestive organs) that are precisely matched to their food sources. The anchoring phenomenon is established: the unit will explain HOW these structures achieve nutrient acquisition. Pedagogical note: record students' initial ideas on the Driving Question Board (DQB) in a 'What we think' column — do NOT correct misconceptions yet; use them as diagnostic data.	Students can explain that despite the visible differences between an aphid, a flamingo, and a human, all three are solving the same biological problem — extracting nutrients from the environment — using structures shaped by their diet. The teacher should use the analogy of tools in a Nairobi hardware shop: a jembe (hoe), a panga, and a pair of pliers all 'harvest' material, but their shapes reflect the specific job. The driving question is posted prominently and the class co-constructs the first DQB entries. Formative check: exit ticket — 'In one sentence, what is animal nutrition, and name ONE structure any animal uses to get food.'
Lesson 2 **Chewing and Biting Mouthparts: How Mandibles Enable Mechanical Breakdown of Solid Food**	Students observed preserved or printed specimens of a locust (Schistocerca gregaria) and a caterpillar (Spodoptera frugiperda — the fall armyworm devastating Kenyan maize). Using hand lenses or a projected diagram, they identified the labrum, mandibles, maxillae, and labium.	Chewing (mandibulate) insects possess mandibles — hardened, sclerotised, toothed cuticular structures that move in the medial plane (side to side) to cut, crush, and grind solid food. The labrum acts as an upper lip to hold food; maxillae and the labium manipulate and taste food. This	Students can explain that the mandible's hardness (from chitin and protein cross-linking), its toothed edges, and its side-to-side motion are structural features that directly enable the function of cutting and grinding solid plant material. This links back to the driving question: the locust and fall

	Teacher should direct: 'Use your hand lens to find the hardest, darkest part of the mouthparts — that is the mandible.' Students then performed a structured observation: they pressed two index fingers together and moved them side-to-side (medially) to simulate mandibular movement, contrasting with up-down jaw movement. This kinesthetic model makes the medial motion concrete. A chewed maize leaf sample and an unchewed leaf are compared under the hand lens.	constitutes mechanical digestion: the physical breakdown of large food particles into smaller ones, increasing surface area for subsequent chemical digestion. The fall armyworm's mandibles are adapted to slice through the tough cellulose of maize leaves. Pedagogical note: emphasise that mechanical digestion does NOT change the chemical identity of food molecules — a starch molecule remains starch whether in a whole grain or a chewed piece.	armyworm devastate Kenyan crops precisely because their mandibles are optimised for shearing plant tissue. Teacher should ask: 'Could a mosquito eat a maize leaf with its mouthparts? Why not?' (Cold-call 3 students; expected answer: no — the mosquito's mouthparts are specialised for piercing and sucking, not chewing.) Add 'mandible structure → mechanical breakdown' to the DQB model column.
Lesson 3 **Piercing and Sucking Mouthparts: How Does a Mosquito Drink Blood but a Grasshopper Cannot?**	Students constructed physical models of piercing-sucking mouthparts using a plastic drinking straw (food canal), a thin coffee stirrer inserted inside (salivary canal), and foam tubing wrapped around (labium). They labelled all parts using masking tape flags. They then used the model to 'pierce' a balloon filled with red-coloured water (representing a blood vessel or phloem sieve tube) and observed liquid rising through the food canal. Teacher should circulate and ask: 'What would happen if the labium were rigid instead of flexible?' (Expected: it could not fold back to allow stylet penetration.) Students also examined a diagram of an aphid stylet navigating between maize leaf cells to reach the phloem — a Kenya-relevant agricultural pest context.	Piercing-sucking mouthparts consist of: (1) the labium — a flexible, grooved sheath that guides the stylet bundle and folds back during feeding; (2) the stylet bundle containing the food canal (through which liquid food is drawn up) and the salivary canal (through which saliva containing anticoagulants or digestive enzymes is injected). The aphid's stylet navigates intercellularly to reach phloem sap; the mosquito's stylet locates a blood capillary. Neither organism can chew solid food. These mouthparts are adapted to liquid diets under pressure. Pedagogical note: the aphid is a major pest on Kenyan bean and maize farms — this context motivates why understanding the structure matters for pest management decisions.	Students can explain that the needle-like, smooth stylet can penetrate tissue because it is thin and rigid (hardened chitin), while the labium's flexibility allows it to concertina without tearing the feeding site. The salivary canal's role in injecting saliva that suppresses plant or host defences is a key structure-function link. Teacher draws the contrast table on the board: mandibulate (locust) vs haustellate (mosquito/aphid) — structures differ because diets differ. Connect to driving question: 'The aphid's stylet explains HOW it extracts phloem sap from a maize plant — the structure IS the explanation.' Update DQB: add 'stylet → liquid feeding' alongside 'mandible → solid food grinding.'
Lesson 4 **Crushing and Tearing Bird Beaks: How Does a Fish Eagle Eat What a Flamingo Cannot?**	Students performed a structured card-sort activity using photograph cards of five Kenyan birds: the African fish eagle (Haliaeetus vocifer), the Rüppell's vulture, the African grey hornbill, the superb starling, and the flamingo. Each card includes a beak image and a food-source clue. Students sorted the cards into groups before receiving any labels. Teacher should give 3 minutes for silent individual sorting, then 2 minutes for pair discussion, then cold-call 3 pairs to justify their groupings. After sorting, students used forceps (simulating a hooked beak) vs. a slotted spoon (simulating a filter beak) to retrieve	Beak morphology in birds is a direct structural adaptation to diet: (1) The African fish eagle's beak is sharply hooked and pointed, with a strong curved upper mandible — adapted to grip slippery fish and tear flesh; the talons complement the beak by securing prey. (2) The vulture's heavily hooked, powerful beak with a long neck allows it to reach deep inside carcasses to tear muscle from bone — a scavenger adaptation seen at Kenyan game reserves. (3) Seed-eaters (hornbill, starling) have short, robust, conical beaks for cracking hard seed coats. The key principle: beak shape = diet fingerprint.	Students can explain that the fish eagle's hooked tip creates a lever that amplifies force at the point of contact, allowing flesh to be torn rather than grasped whole. The curvature of the upper mandible means the bird can 'hook and pull' — a structure matched exactly to the function of dismembering a Nile perch caught from Lake Victoria. Teacher should ask: 'Why can't the fish eagle filter Lake Nakuru's algae like a flamingo?' (Answer: its beak has no lamellae/filter plates, and the hook would be useless for concentrating microscopic algae.) This contrast sets up Lesson 5 perfectly. Add to DQB: 'hooked

	different items from a water tray: rubber fish shapes, small beads, and larger pebbles — making beak function tactile.	Pedagogical note: 'You can identify what a bird eats just by looking at its beak' is a powerful takeaway — encourage students to test this the next time they see a bird near school.	beak → gripping/tearing flesh.'
Lesson 5 **Probing and Filtering Bird Beaks: How Beak Morphology Reveals Diet and Habitat**	Students simulated flamingo filter-feeding using a bent plastic spoon with holes drilled or cut into it (representing lamellae). They held it upside down in a tray of water containing small sesame seeds mixed with larger lentils, moved the 'beak' side-to-side while submerged, then observed which seeds were trapped. Teacher should ask: 'Why does the flamingo hold its beak upside down?' (Allow 30 seconds think time; cold-call 2 students.) Students also examined a close-up photograph of flamingo lamellae and compared it to a photograph of a whale's baleen — drawing the parallel between two very different animals using the same filtration principle. A Kenya context anchor: the 1–2 million lesser flamingos at Lake Nakuru represent one of the greatest wildlife spectacles, sustained entirely by the filtration of Spirulina algae.	The flamingo's beak is uniquely bent at 90° at the midpoint, held upside down during feeding, and lined with fine comb-like lamellae on the upper mandible and a large, muscular tongue that acts as a pump. Water and algae are taken in; the tongue pumps water out through the lamellae filter while algae and diatoms are trapped and swallowed. The sunbird's long, curved, tubular beak is adapted for nectar-probing in deep flowers (e.g., aloe species common in Kenyan highlands). The key concept: filtering structures select for particle SIZE, not species — lamellae are a physical sieve. Pedagogical note: Compare to the fine mesh on a Kenyan tea strainer vs. a colander — same principle, different pore size, different selectivity.	Students can explain that the flamingo's upside-down beak position, the muscular tongue pump, and the fine lamellae together constitute a biological filtration system precisely calibrated to the size of Spirulina algae cells (~5–10 µm). This explains why flamingos concentrate at Lake Nakuru (alkaline lake with dense Spirulina blooms) rather than Lake Victoria (different algae profile). The driving question is revisited: 'The flamingo's beak structure — its bend, its lamellae, its tongue — directly explains HOW it extracts nutrients (cyanobacteria) from Lake Nakuru.' Update DQB: students add 'filter beak + lamellae → microscopic algae extraction.' Class now has insect mouthparts AND bird beak examples on the board.
Lesson 6 **Dentition in Mammals: Tooth Shape, Diet, and the Structure-Function Connection**	Students examined plaster cast models or printed photographs of three mammalian skulls: a lion (Panthera leo, carnivore), a zebra (Equus quagga, herbivore), and a human (Homo sapiens, omnivore). Using a structured observation sheet, they counted and sketched tooth types in each quadrant. Teacher should circulate with the prompt: 'Find the longest, most pointed tooth in each skull — what job do you think it does?' Students also bit into a piece of ugali (soft, cohesive starchy food) and noted which teeth they used first (incisors to cut), then chewed with (molars). This self-observation connects their own dentition directly to the lesson content.	Mammals possess four types of teeth, each structurally adapted to a specific function: (1) Incisors — flat, chisel-shaped, at the front; cut and slice food (e.g., biting a piece of chapati). (2) Canines — long, conical, pointed; pierce and hold prey (large in carnivores like lions; reduced in herbivores). (3) Premolars — broader, with cusps; crush and begin grinding. (4) Molars — large, flat, multi-cusped; grind tough fibrous material (largest in herbivores). The dental formula records tooth numbers per half-jaw (I/I · C/C · P/P · M/M × 2). Human dental formula: I 2/2 · C 1/1 · P 2/2 · M 3/3 × 2 = 32 teeth. Pedagogical note: students commonly confuse 'dental formula is per half-jaw × 2' — emphasise this explicitly and have them calculate the human total (32) step by step.	Students can explain that a lion's reduced/absent premolars and huge canines reflect a diet of fresh meat requiring gripping and shearing, while the zebra's absent canines and vast flat molars reflect a diet of tough savanna grasses requiring prolonged grinding. The human omnivore dentition is a 'compromise' — functional incisors, small canines, and effective molars — matching a diet of diverse foods from nyama choma to githeri to fruit. Teacher should ask: 'If you found a mammal skull with no canines and very large molars covering most of the jaw, what would you predict about its diet? Where in Kenya might it live?' (Expected: herbivore — perhaps a wildebeest on the Maasai Mara.) Add to DQB: 'tooth shape → diet type → habitat prediction.'
Lesson 7	Students used a 7-metre length of rope laid	The human digestive system is a	Students can explain that the sequence of

The Human Digestive System: Tracing a Meal of Ugali and Sukuma Wiki	out on the school compound to represent the full length of the human alimentary canal — mouth to anus. Different coloured ribbon sections mark each organ (red = stomach, yellow = small intestine ~6 m, green = large intestine ~1.5 m). Students walked the length of the rope while the teacher narrated: 'Your ugali just left your mouth… it's now in your oesophagus… it just entered your stomach.' This kinaesthetic activity makes the scale of the digestive system viscerally real. Students then completed a blank diagram labelling activity identifying: mouth, salivary glands, oesophagus, stomach, liver, pancreas, small intestine (duodenum, ileum), large intestine, rectum, and anus.	continuous tube (the alimentary canal) plus accessory glands. Each organ has a structure matched to its function: (1) Mouth — teeth and tongue for mechanical digestion; salivary glands secrete amylase for starch breakdown. (2) Oesophagus — muscular tube using peristalsis (wave-like muscle contractions) to move food. (3) Stomach — thick muscular walls churn food into chyme; gastric glands secrete HCl (pH 2, kills pathogens, activates pepsin) and pepsin (protein digestion). (4) Small intestine — main site of chemical digestion and absorption; receives bile from liver (emulsifies fats) and pancreatic enzymes. (5) Large intestine — absorbs water; forms faeces. Pedagogical note: the 6-metre length of the small intestine surprises students — use this surprise to motivate the question: 'Why so long?' (Answer explored in Lesson 9: maximum absorption surface area.)	organs represents a production line: each station (organ) performs a specific transformation on the ugali-and-sukuma-wiki meal. The thick muscular stomach wall produces the mechanical churning needed to mix the meal with gastric acid, while the long small intestine provides the time and surface area needed for absorption. Teacher connects to the driving question: 'Just as the flamingo's beak is a structure that explains HOW it feeds, the stomach's thick muscle wall is a structure that explains HOW it breaks down your ugali.' Update DQB: students add organ names to their cumulative model diagram of the digestive system.
Lesson 8 **Enzymes in Digestion: How Ugali and Sukuma Wiki Are Chemically Broken Down**	Students performed a starch-amylase experiment: they placed a drop of cooked ugali paste (starch suspension) in each well of a spotting tile, added a drop of salivary amylase solution (or commercial amylase), and tested with iodine solution at 30-second intervals. Iodine turns blue-black in the presence of starch; as amylase digests the starch the blue-black colour fades to orange-brown. Students recorded colour changes on a time table. Teacher should ask: 'What is the iodine test telling you is DISAPPEARING?' (Starch.) 'What is APPEARING instead?' (Reducing sugars — confirmed with Benedict's reagent on a final sample.) This two-test protocol makes both substrate disappearance and product appearance visible. Temperature variation (cold water vs. 37°C water bath) can be added as an extension.	Chemical digestion is the enzymatic hydrolysis of large, insoluble food molecules into small, soluble molecules that can be absorbed. Key enzymes and their actions: (1) Salivary amylase (mouth) — breaks starch → maltose (a reducing sugar). (2) Gastric protease/pepsin (stomach, active at pH 2) — breaks proteins → polypeptides. (3) Pancreatic amylase (small intestine) — continues starch → maltose digestion. (4) Pancreatic lipase (small intestine) — breaks lipids → fatty acids + glycerol (bile salts from the liver first emulsify fat droplets, increasing surface area for lipase). (5) Intestinal proteases (small intestine) — complete protein → amino acid digestion. Pedagogical note: a common misconception is that enzymes are 'used up' — explicitly state that enzymes are biological catalysts; they are reused. Use the analogy of a mkokoteni (handcart) that carries goods without being consumed.	Students can explain that the ugali (maize starch and protein) is chemically dismantled in a relay: amylase begins in the mouth, is inactivated by stomach acid, and is resumed by pancreatic amylase in the small intestine. Sukuma wiki (kale — containing cellulose, vitamins, and minerals) has its cell walls mechanically disrupted by chewing; vitamins and minerals are released for absorption, but cellulose is NOT digested (humans lack cellulase) and passes to the large intestine as dietary fibre. Teacher should ask: 'Why does your saliva start digesting ugali but not nyama choma (meat)?' (Salivary amylase acts on starch, not protein — correct enzyme for correct substrate.) Add to DQB: 'enzyme specificity → substrate-specific chemical breakdown.'
Lesson 9	Students compared two surfaces to understand absorption area: a flat plastic	Absorption occurs primarily in the ileum. Structural adaptations maximising	Students can explain that the ileum's villi and microvilli are structural adaptations that

Absorption, Assimilation, and Malnutrition: From Villi to Village Health	sheet vs. a towel made of terry-cloth loops. They pressed both against a wet paper towel for 5 seconds and compared water absorbed — the towel (representing villi) absorbs far more. Teacher then shows a photomicrograph of ileum cross-section displaying villi and microvilli, alongside a labelled diagram. Students measured and calculated: 'If one villus increases surface area by 10×, and microvilli increase it by 20× more, what is the total multiplication factor?' (200×.) Students also analysed a case study: a 12-year-old child from a rural Kisumu family whose diet consists mainly of plain ugali with no protein or vitamin sources — students identify which nutrients are missing and predict health consequences.	absorption: (1) Villi — finger-like projections of the intestinal lining increasing surface area. (2) Microvilli (brush border) — further increase surface area on each villus epithelial cell. (3) Rich capillary network in each villus — glucose and amino acids diffuse into blood capillaries → hepatic portal vein → liver. (4) Lacteals — lymph vessels in each villus absorb fatty acids and glycerol (reassembled into triglycerides as chylomicrons). Assimilation = the use of absorbed nutrients by body cells (e.g., glucose → ATP in cellular respiration; amino acids → new proteins for muscle growth). Malnutrition in Kenya: protein deficiency → kwashiorkor (oedema, stunted growth); energy deficiency → marasmus; iron deficiency → anaemia (prevalent in Kisumu and Homa Bay counties). Pedagogical note: handle malnutrition content sensitively — frame around systemic food access issues, not individual blame.	increase surface area to maximise the rate of absorption — exactly the same logic as the flamingo's lamellae increasing the surface area for filtering algae. The structure-function connection is identical in principle, applied to different scales and contexts. Teacher draws this explicit parallel on the board. The Kisumu case study is debriefed: the child lacks protein (no amino acids for growth → kwashiorkor risk), iron (no haemoglobin synthesis → anaemia risk), and vitamins. Update DQB: 'villus structure → maximum absorption surface area.' Students revise their cumulative model to include absorption and assimilation steps.
Lesson 10 **Synthesis and Model Finalisation: From Aphid Stylets to Human Villi — One Purpose, Many Structures**	Students laid out all anchor images from the unit (aphid on maize, mosquito, locust, flamingo at Lake Nakuru, fish eagle over Lake Victoria, lion skull, zebra skull, human digestive system diagram) on their desks simultaneously. They were given 5 minutes to silently re-examine all images with the driving question projected: 'How do the structures animals use to get food explain how each animal gets the nutrients it needs to survive?' Teacher then posed three synthesis questions for think-pair-share: (1) 'What is the ONE biological problem all these structures are solving?' (2) 'Choose any two animals from the unit — explain how their feeding structures are similar in PURPOSE but different in FORM.' (3) 'Add one animal not studied in this unit and predict its feeding structure based on what you now know.' Cold-call 4 pairs for question 3.	The unifying principle of the entire unit: all animal feeding structures — regardless of their physical form — are evolutionary solutions to the same problem: extracting sufficient nutrients (energy, protein, minerals, vitamins) from the environment to support life. Structure always reflects function, and function always reflects diet/habitat. Specific structures: aphid stylet (liquid phloem sap, intercellular penetration), mosquito stylet (liquid blood, anticoagulant saliva), locust mandibles (solid plant tissue, mechanical shearing), flamingo lamellae (microscopic algae, pressure filtration), fish eagle hooked beak (fish flesh, gripping and tearing), mammalian dentition (diet-matched tooth morphology), human alimentary canal (sequential mechanical + chemical digestion + absorption + assimilation). Pedagogical note: this synthesis lesson is the payoff of all prior lessons — ensure every student can articulate the driving question answer in their own words before	Students can fully explain the driving question: the structures animals use to get food are not arbitrary — each feature (the needle sharpness of a stylet, the hook curvature of an eagle's beak, the flat surface of a molar, the finger-like projection of a villus) is a precise structural adaptation that makes nutrient extraction possible for that animal in that environment. Form follows function follows diet follows habitat. Teacher conducts a final DQB review: read each entry from Lessons 1–9 and ask, 'Do we now have an explanation for this?' Students vote thumbs-up if the entry is now explained. Any unresolved questions are noted for future study. Students submit their final annotated cumulative model as the unit's summative performance task — the model must include at least one insect mouthpart, one bird beak, one mammalian tooth type, and one labelled digestive organ, each linked by arrows to the nutrient it helps obtain.

		the lesson ends.	